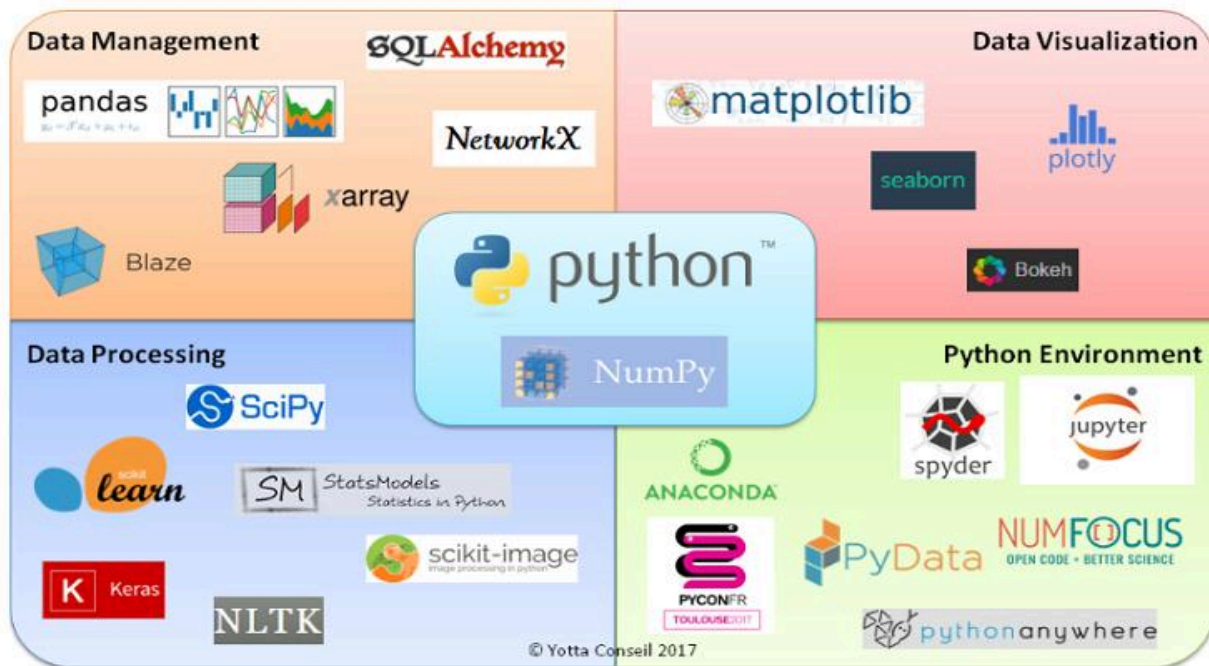


583- Introduction to python



2441 - Why Python for Data/AI

Python is a high-level programming language widely used in Data Science, Machine Learning, Artificial Intelligence, and Automation because of its simplicity and powerful libraries.

- Easy to learn and read.
- Open-source and free to use.
- Supports Data Science and AI applications.
- Has powerful libraries such as NumPy, Pandas, and Scikit-Learn.
- Large community and industry adoption.

2442 - Install Python

Python installation is the process of downloading and setting up Python on a computer so that Python programs can be written and executed.

Steps

1. Download Python from the official website.

2. Run the installer.
3. Select **Add Python to PATH**.
4. Click **Install Now**.
5. Open Command Prompt (Windows) or Terminal (Mac/Linux).
6. Verify the installation.

```
python --version
```

2443 - Development Environments

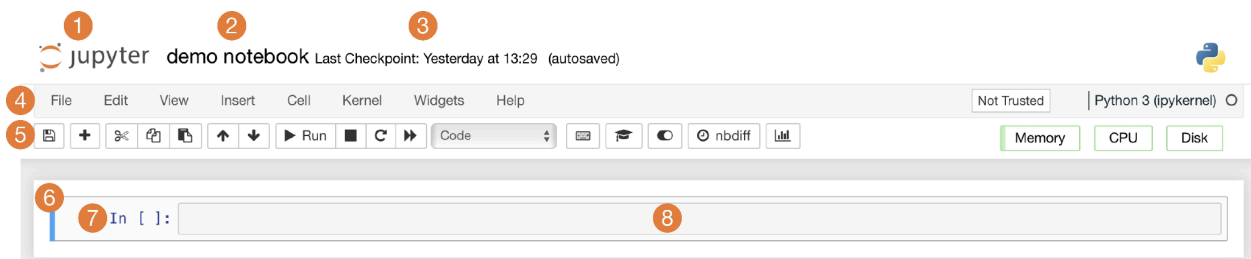
A Development Environment is a software application or workspace used to write, run, test, and debug Python programs.

Common Development Environments

- **IDLE** – Default Python editor.
- **VS Code** – Lightweight and popular code editor.
- **PyCharm** – Professional Python IDE.
- **Jupyter Notebook** – Interactive environment for Data Science and Machine Learning.

2444 - Jupyter Notebook

Jupyter Notebook is an interactive web-based application that allows users to write, run, and document Python code in individual cells. It is widely used in Data Science, Machine Learning, and Data Analysis.



Steps

1. Install Anaconda.
2. Launch Jupyter Notebook.
3. Create a new notebook.

4. Add code cells and run them using **Shift + Enter**.

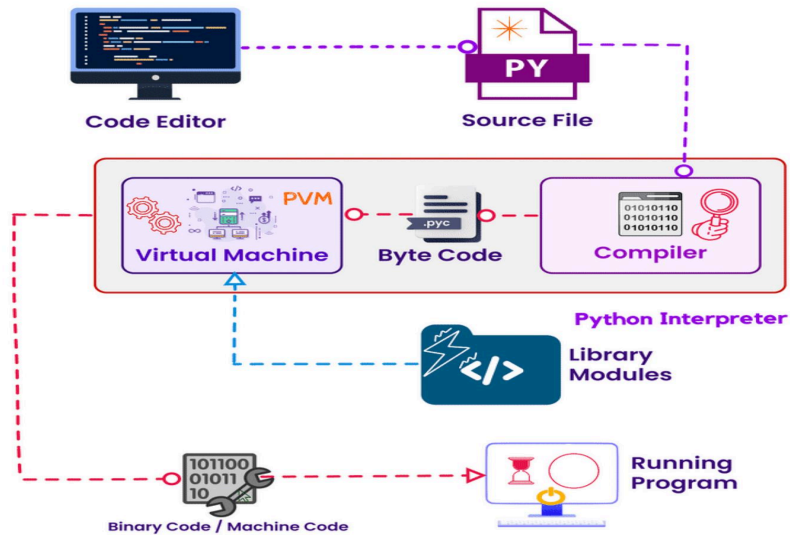
2445 - Script vs Notebook

A Script is a Python file (.py) that contains code executed as a complete program. A Notebook is a Jupyter file (.ipynb) where code is executed one cell at a time.



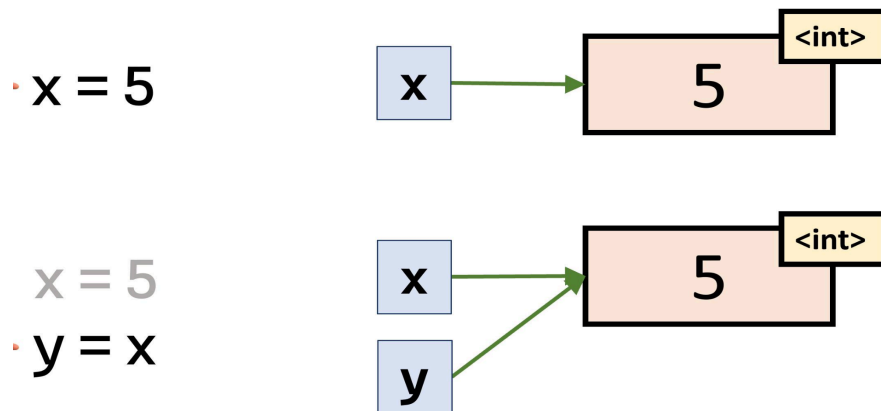
2446 - Execution Model

The Python Execution Model describes how Python processes and executes code to produce the desired output.



2447 - Memory Layout

Memory Layout refers to how Python stores variables and objects in memory during program execution.



584- Variables & Types

2448 - Variables

A variable is a named container used to store data values in a program.

Assigning the value
to the variable

first_name = "John"

Variable name Value

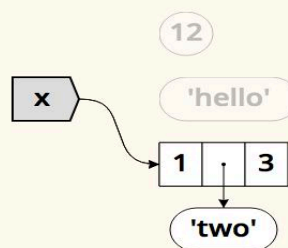
2449 - Dynamic Typing

Dynamic Typing is a feature in Python where the data type of a variable is determined automatically at runtime and can change during program execution.

Any name → any value @ any time

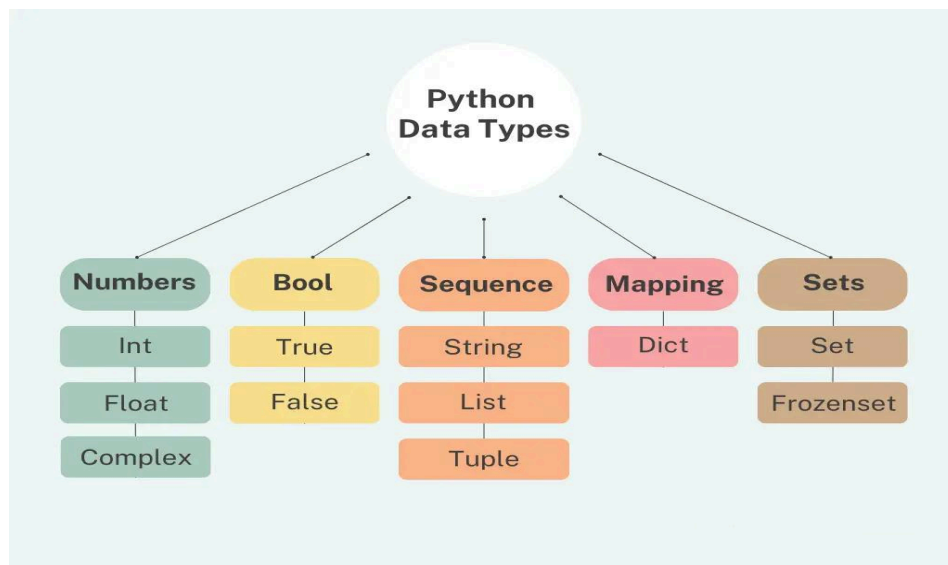
Dynamic typing

```
x = 12  
x = "hello"  
x = [1, 2, 3]  
x[1] = "two"
```



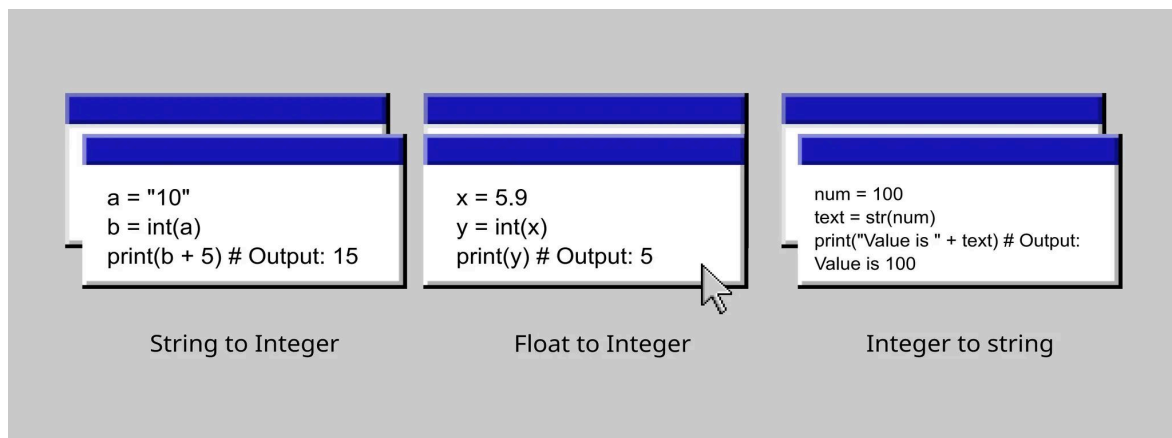
2450 - Data Types

Data Types define the type of data a variable can store, such as numbers, text, or logical values.



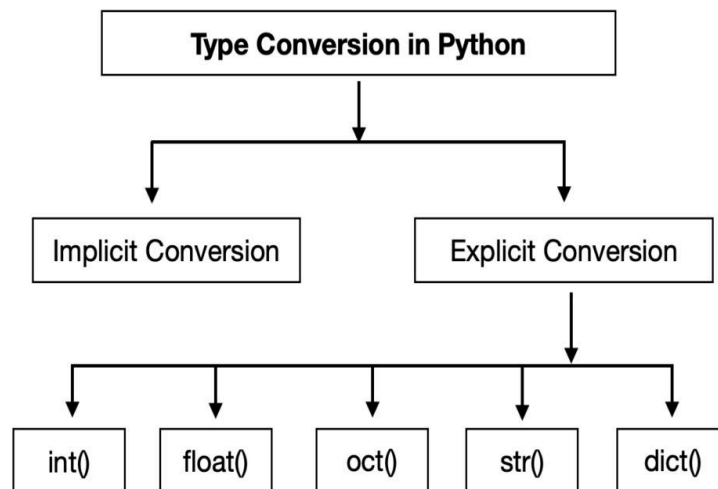
2451 - Type Casting

Type Casting is the process of converting a value from one data type to another using built-in Python functions such as `int()`, `float()`, `str()`, and `bool()`.



2452 - Type Conversion

Type Conversion is the automatic or manual process of changing a value from one data type to another compatible data type.



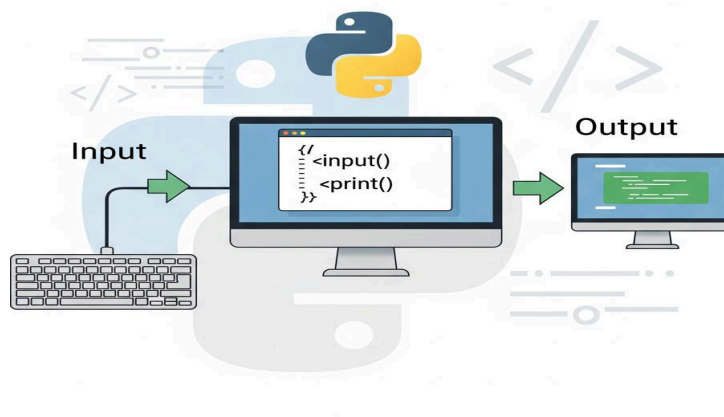
585:Input/Output

2453 - Input

Input is the data or information provided to a computer or program for processing. Input can come from users, files, databases, sensors, or other systems.

2454 - Output

Output is the information or result produced by a computer or program after processing the input data.



586- String Formating

2455 - F-String

F-String (Formatted String Literal) is a feature in Python used to insert variables and expressions directly inside a string by prefixing the string with `f`.

```
temperature = 25
weather_condition = "sunny"

message = f"The current temperature is {temperature}°C, and the weather is {weather_condition}."
print(message)
```

587- Comments

2456 - Comments

Comments are non-executable lines in a Python program used to explain code, improve readability, and provide notes for developers.


```
# Single line comment
```

```
print('Hello Stechies')
```

Single-line Comment

```
'''  
Python program to explain  
multiple line comment  
print('Hello')  
'''
```

Multi-line Comments

```
"""  
Python program to explain  
multiple line comment  
print('Hello')  
"""
```

```
print('Hello Stechies')
```

588-Indentation

2457 - Indentation

Indentation is the use of spaces or tabs at the beginning of a line of code to define code blocks in Python. It determines the structure and flow of a program.

Python uses Indentation

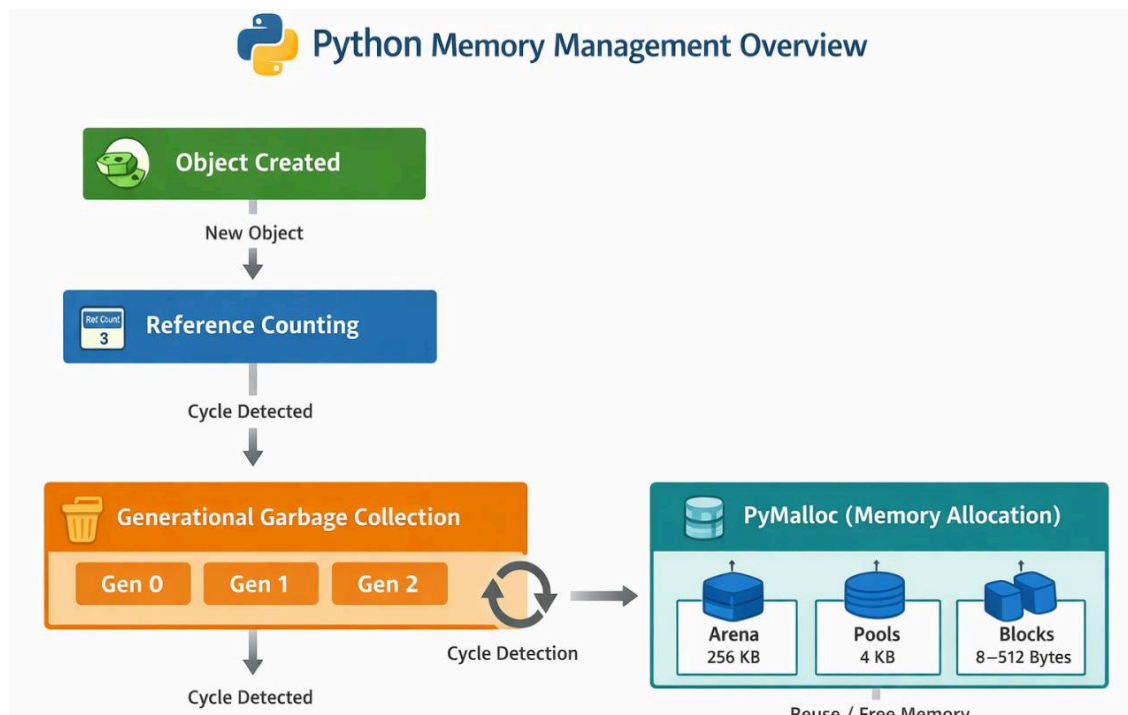
```
if a==1:  
    print(a)  
    if b==2:  
        print(b)  
print('end')
```



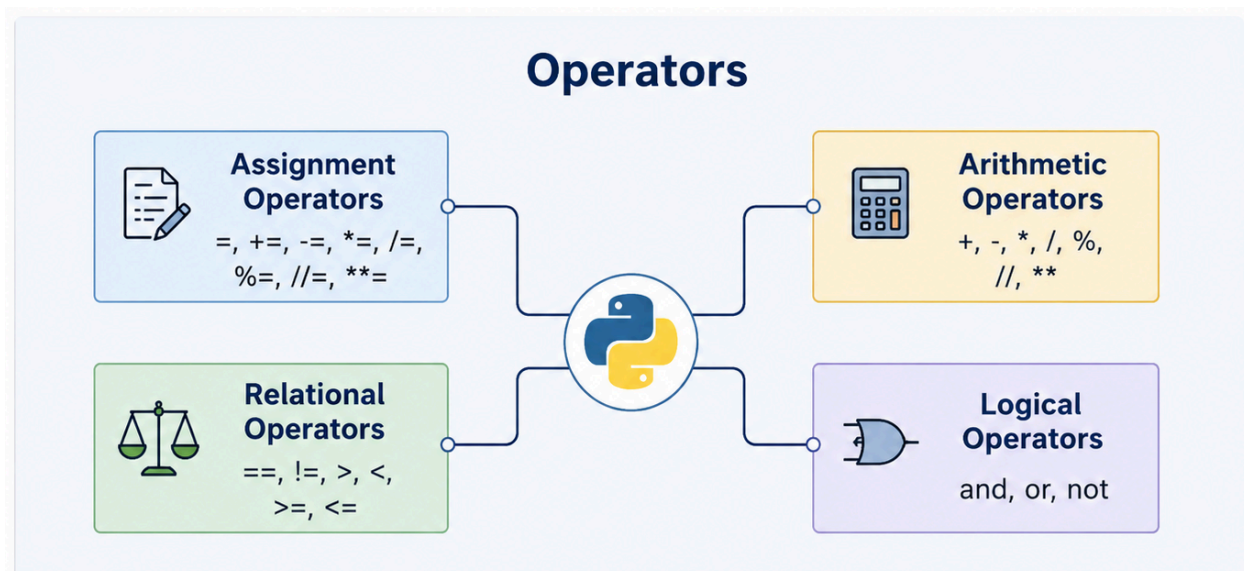
589 Memory Management

2458 - Memory Management

Memory Management is the process by which Python allocates, uses, and releases memory automatically during program execution.



560 - Operators



2459 - Assignment Operators

Assignment Operators are used to assign values to variables. They can also perform operations and assign the result to the same variable.

2460 - Logical Operators

Logical Operators are used to combine or evaluate multiple conditions and return either True or False.

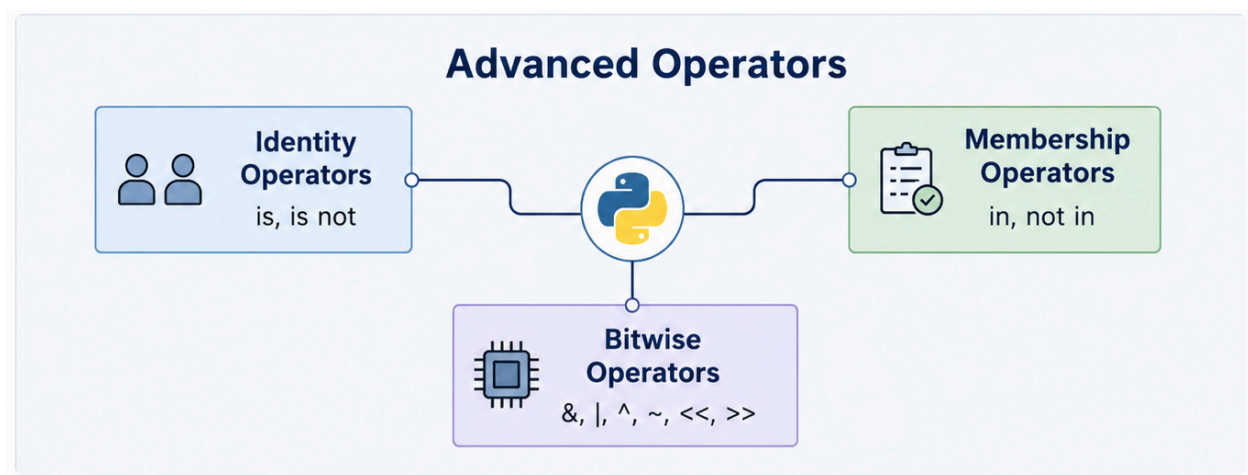
2461 - Relational Operators

Relational Operators are used to compare two values and determine the relationship between them, such as equal to, greater than, or less than.

2462 - Arithmetic Operators

Arithmetic Operators are used to perform mathematical calculations such as addition, subtraction, multiplication, division, and modulus operations.

561 - Advanced Operators



2463 - Identity Operators